

 HARIGANESH_241801078 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
[HARIGANESH_241801078@localhost ~]$ gcc producer_consumer.c  
[HARIGANESH_241801078@localhost ~]$ gcc producer_consumer.c  
[HARIGANESH_241801078@localhost ~]$ ./a.out
```

```
1. Produce  
2. Consume  
3. Exit  
Enter choice: 1  
Enter item: 1  
Produced: 1
```

```
1. Produce  
2. Consume  
3. Exit  
Enter choice: 1  
Enter item: 2  
Produced: 2
```

```
1. Produce  
2. Consume  
3. Exit  
Enter choice: 2  
Consumed: 1
```

```
1. Produce  
2. Consume  
3. Exit  
Enter choice: 3  
[HARIGANESH_241801078@localhost ~]$
```

```
#include <stdio.h>
#include <stdlib.h>

#define SIZE 5

int buffer[SIZE];
int in = 0, out = 0, count = 0;

void produce(int item) {
    if (count == SIZE) {
        printf("Buffer FULL! Cannot produce\n");
        return;
    }
    buffer[in] = item;
    in = (in + 1) % SIZE;
    count++;
    printf("Produced: %d\n", item);
}

void consume() {
    if (count == 0) {
        printf("Buffer EMPTY! Cannot consume\n");
        return;
    }
    int item = buffer[out];
    out = (out + 1) % SIZE;
    count--;
    printf("Consumed: %d\n", item);
}

int main()
-- INSERT --
```

```
int main() {
    int choice, item;

    while (1) {
        printf("\n1. Produce\n2. Consume\n3. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter item: ");
                scanf("%d", &item);
                produce(item);
                break;
            case 2:
                consume();
                break;
            case 3:
                exit(0);
                break;
            default:
                printf("Invalid choice!\n");
        }

        return 0;
    }
}

-- INSERT --
```